Vyom Vyas

vyom2512vyas@gmail.com 9428336004 LinkedIn Github

EDUCATION

Bachelors of Technology (B.Tech), Electrical Engineering, **2022 – 2026**
Sardar Vallabhbhai National Institute of Technology, Surat

PROFESSIONAL EXPERIENCE

ADAS Developer, *PEM Technologies* **Jul 2026 – Present** | *Bangalore, India*

- Developed prediction, planning and control for the company's AD stack, with an independent **AEB** layer and a **gRPC** link to **CARLA**, validated closed-loop at **20 Hz** on AEB, cut-in, vulnerable-road-user (VRU) and occlusion scenarios.
- Developing a scenario concretisation prototype that turns **PEGASUS**-style logical scenarios (parameter ranges) into concrete, physically consistent test cases for simulation.
- Defining per-case avoidability labels from **UN R157**, **ISO 34502** and Euro NCAP references.

AD Platform Intern, *PEM Technologies* **Jan 2026 – Jun 2026** | *Bangalore, India*

- Built the **ODD**-based scenario generation pipeline (Python, **FastAPI**, **PostgreSQL**): from **145** ODD elements, removed about **48%** of combinations as physically infeasible and wrote the rest up as **PEGASUS** logical scenarios built around edge-case **SOTIF** triggering conditions.
- Wrote early planning and control for the AD stack in **CARLA**: path tracking, lane changes, obstacle avoidance with replanning; built its KPI logging.
- Built a **3D LiDAR SLAM** and navigation demo for off-road vehicles (**ROS 2**, **Gazebo**) for a customer demo.

Intern - Autonomous Mobile Robotics, *Novus HiTech Robotics Systems* **May 2025 – Jul 2025** | *Gurugram, India*

- Tested and tuned an **Extended Kalman Filter (EKF)** in **roscpp** fusing wheel encoder and IMU data, with **Madgwick** and 3rd-order low-pass filtering of IMU noise.
- Measured **2 cm/m** odometry drift, down from **15 cm/m**, and **85–90%** accuracy in **Gazebo** and on the warehouse robot.

SKILLS

- **AD Frameworks:** CARLA, Autoware, ROS/ROS2.
- **Robotics & Perception:** LiDAR, SLAM (EKF and Graph based, GMapping), Gazebo, RViz, Nav2, 3D-SLAM.
- **Navigation & Control:** Path Planning (Hybrid A*), Motion Planning, Motion Prediction, Vehicle Dynamics, Path Tracking Control (PID, Pure Pursuit, Stanley, LQR).
- **Testing & Safety:** Scenario-based Testing, Verification & Validation, SOTIF (ISO 21448), UN R157 / R152, Euro NCAP.
- **AI & Computer Vision:** PyTorch, Deep Learning, Reinforcement Learning, YOLO, CNNs, LSTMs, VAEs.
- **Programming & Tools:** Python, C++, MATLAB, Docker, Linux, Simulink, Git, gRPC

PROJECTS

CARLA-MATLAB Control Bench, Path tracking controller benchmark | MATLAB, Simulink, CARLA GitHub

- Compared **PID**, **Pure Pursuit**, **Stanley** and **LQR** steering controllers in **CARLA** over **4** scenarios: a 174 m radius highway curve, a 4.9 m junction turn, a hairpin, and a double lane change.
- **LQR** (bicycle model, per-speed Riccati gain schedule) tracked best on all **4**, holding **0.031 m** RMS cross-track error through the double lane change at **160 km/h**, where fixed-gain PID drifted to **2.2 m**.
- Added curvature based speed profiling (braking from **120 km/h** into the 4.9 m turn) and ran the same controllers from a **Simulink** model over **UDP** at **20 Hz**.

Pose Graph SLAM, 2D LiDAR and Odometry Pose Graph SLAM from scratch | ROS2 - C++ GitHub

- Deployed a **ROS2** pipeline in **C++**, implementing **ICP scan matching** using 2D LiDAR for loop closure detection across **668 pose nodes** with **24 loop closures**.
- Attained **99.94% chi^2 error reduction** (11,404 to 6.79) in **7 Gauss-Newton iterations** on TurtleBot3 in Gazebo, correcting drift from artificial odometry noise.

EKF-SLAM, Simultaneous Localization and Mapping using Odometry & LiDAR from scratch GitHub

- Programmed an **Extended Kalman Filter (EKF)** based SLAM framework from scratch to fuse wheel odometry and 2D LiDAR data for joint pose and landmark map estimation.
- Formulated a custom **2 layered nested-loop matcher** to align LiDAR edge features with cylindrical landmarks and maintain a consistent global map in real time.
- Visualized trajectories, uncertainty ellipses, and landmarks in **RViz2**, benchmarking **0.002 m** pose accuracy in Gazebo.

ORION- Rover, Autonomous all-terrain drive robot Image

- Built **rclpy** and **rclcpp** nodes for 2D LiDAR, GPS, IMU and webcam navigation, verified in Gazebo.
- Fused **YOLOv8n** detections with 2D LiDAR depth estimates for real-time arrow recognition, alongside a **6-DOF robotic arm** teleoperation pipeline.

ACHIEVEMENTS

GUJCOST Robofest 3.0 - Gujarat government -2023, RCL-E Rover Video

- Played a role in autonomy development that helped secure **1st place** at GUJCOST Robofest 3.0, outperforming top IITs/NITs.

International Rover Challenge, Space Robotics Society Jan 24 & 25 Certificate

- Earned **14th place** (2025) and **17th place** (2024); **Top 10** finalist, Rover Design Challenge '25.